

Quality Management

5 — *ISO 9000 and ISO 9001*

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Today

Last time ended with a warning

We finished with the quality manual — and the observation that a manual nobody reads is a filing cabinet, not a quality system.

ISO 9001 is the answer to that problem. It turns the ideas of the last two lectures into requirements that an outsider can audit, and that a certificate can attest to.

1

The family

what ISO 9000 is, and how it differs from 9001

2

Seven principles

the ideas the whole family rests on

3

ISO 9001

the structure, clause by clause

PART 1

The ISO 9000 family

*Four standards, and one distinction students get
wrong every year*

ISO 9000 is not ISO 9001

Get this right and the rest of the lecture is easy

ISO 9000 — fundamentals and vocabulary

The **foundation**. It defines the concepts, the principles and the terminology. It gives everyone a common language for quality. It contains *no requirements*, so **you cannot be certified against it**.

ISO 9001 — requirements

The **certification standard**. It takes the concepts of ISO 9000 and turns them into specific, auditable requirements for a quality management system. This is what an organisation is certified against.

THE DICTIONARY AND THE DRIVING TEST

ISO 9000 is the dictionary: it tells you what the words mean. ISO 9001 is the driving test: it tells you what you must actually do, and someone watches you do it. Nobody is ever “certified in the dictionary”.

The phrase to avoid

“We are ISO 9000 certified” is heard constantly in industry and is strictly wrong. The certificate says **ISO 9001**.

Four standards, four jobs

- **ISO 9000** — fundamentals and vocabulary. The concepts and terminology everything else uses.
- **ISO 9001** — requirements for a quality management system. The one you certify against.
- **ISO 9004** — guidance on sustained success. How to go *beyond* the minimum requirements toward long-term performance. Guidance, not requirements — so again, not certifiable.
- **ISO 19011** — guidelines for auditing management systems. How the checking itself is done.

Only one of the four is certifiable

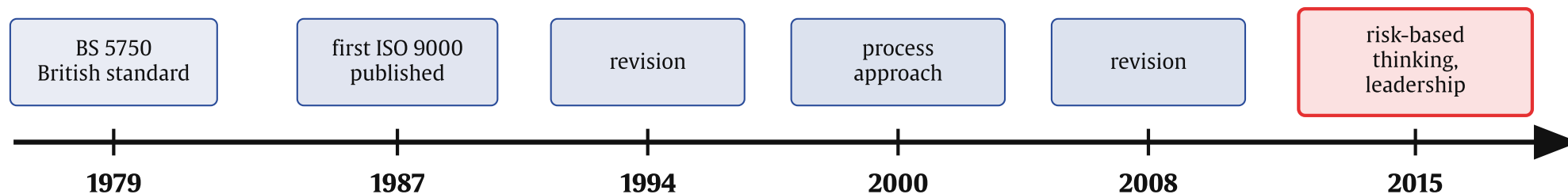
ISO 9001. The other three are vocabulary, guidance and audit method. That asymmetry is the shape of the whole family.

WHY 9004 EXISTS AT ALL

ISO 9001 is a floor, not a ceiling — it describes the minimum that earns a certificate. An organisation that has satisfied it and wants to keep improving has nothing left to conform to. ISO 9004 is written for that organisation.

Where it came from

British military procurement, by way of manufacturing, to everything



Born in manufacturing; now used in healthcare, education and services alike.

WHY A MILITARY STANDARD BECAME A WORLD STANDARD

BS 5750 existed so the British Ministry of Defence could tell whether a supplier was capable *before* placing an order. That is still exactly what ISO 9001 does — the customer just changed.

How ISO defines the words

Compare these with the definitions from Lecture 1

- **Quality** — the **degree** to which a set of inherent characteristics fulfils requirements.
- **Quality management system** — coordinated activities to direct and control an organisation with regard to quality.
- **Non-conformity** — non-fulfilment of a requirement.

NOTICE THE WORD “DEGREE”

In Lecture 1 quality was “conformance to requirements” — a yes or no. ISO 9000 says *degree*, which makes it a scale. That is not pedantry: a scale can be measured, trended and improved. A yes/no can only be passed or failed.

NON-CONFORMITY IS DELIBERATELY NEUTRAL

It is not “defect”, not “failure”, not “fault”. A requirement was not met — that is all it claims. The neutrality matters, because an auditor who writes “defect” has assigned blame before the investigation.

- **1 ISO 9000 is vocabulary; ISO 9001 is requirements.** Only 9001 is certifiable — “ISO 9000 certified” is wrong.
- **2 9004 is guidance beyond the minimum; 19011 says how to audit.** Three of the four contain no requirements at all.
- **3 From BS 5750 in 1979,** through 1987, 1994, 2000 and 2008, to the 2015 edition in force today.
- **4 ISO says “degree to which”** — making quality a scale you can trend, not a pass/fail.

PART 2

The seven principles

*Every requirement in ISO 9001 traces back to one of
these*

Customer focus & Leadership

Principles 1 and 2

- **Customer focus** — meeting and exceeding customer expectations, and improving satisfaction continuously.
- **Leadership** — unity of purpose and direction, and an environment where people are engaged in achieving the objectives.

WHO IS THE CUSTOMER OF A UNIVERSITY?

The student pays and attends. The employer hires the graduate. The state funds the place. All three are customers, all three have different requirements, and they conflict — the employer wants a narrow specialist, the state wants broad employability, the student wants both plus a life. *Naming your customers is the hard part; satisfying them is downstream of that.*

THE LEADERSHIP TEST

If the managing director has never read the quality policy, neither will anyone else. This is why the 2015 revision removed the “management representative” — delegating quality to one person was how leadership was avoided.

Engagement of people & Process approach

Principles 3 and 4

- **Engagement of people** — competent, empowered people at every level, with the skills and resources to contribute.
- **Process approach** — managing activities as interconnected processes, and understanding how they interact.

THE OPERATOR ALREADY KNOWS

Somebody on that line knows why the machine jams on Monday mornings. They have known for two years. Nobody has asked, because Taylor's split said planning was not their job. Engagement is the deliberate undoing of that — and it is free information.

DEFECTS LIVE AT THE HANDOVERS

Most problems do not occur inside a step; they occur *between* steps — shift to shift, department to department, supplier to line. Each department can be individually excellent and the product still fails. That is precisely why you manage processes rather than functions.

Improvement & Evidence-based decision making

Principles 5 and 6

- **Improvement** — continuous improvement as a permanent objective, driven by feedback and data.
- **Evidence-based decision making** — decisions grounded in the analysis of reliable data.

“THE LINE FEELS SLOW LATELY”

That is not evidence. The cycle-time distribution for the last 400 parts is. And notice: once you have the distribution, Shewhart’s question returns — is this drift a special cause worth chasing, or the process being itself?

IMPROVEMENT MUST BE PERMANENT, NOT A PROJECT

An improvement programme with an end date is a project. The principle asks for something harder: that the organisation still behaves this way in year five, when the consultants have gone and the original champion has been promoted.

Relationship management

Principle 7 — and where you have met all seven before

- **Relationship management** — managing relationships with interested parties — suppliers, partners, customers — to create mutual value.

THE SQUEEZE THAT COSTS MONEY

Squeezing a single-source supplier on price until they cut their own inspection does not save money. It moves the defect from their factory to yours, where it is worth ten times more. Their quality system is part of your quality system.

Nothing here is new

Customer focus is Juran's *fitness for use*. Engagement of people is Ishikawa's *quality circle*. Improvement is Deming's *PDCA*. Process approach is the *client–supplier chain*. ISO 9001 did not invent any of this — it wrote down what Japan had already proved, in a form an auditor can check.

- **1 Customer focus & Leadership** — naming your customers is the hard part; and if the top does not read the policy, nobody will.
- **2 Engagement of people & Process approach** — the operator already knows, and defects live at the handovers between steps.
- **3 Improvement & Evidence-based decisions** — permanent, not a project; and a feeling is not data.
- **4 Relationship management** — your supplier's quality system is part of yours.
- **5 All seven are older than the standard.** ISO wrote down what Deming, Juran and Ishikawa had already demonstrated.

PART 3

ISO 9001

*The requirements, the structure, and what a certificate
really says*

What ISO 9001 is

- The standard for **establishing a quality management system** — the most widely recognised in the family.
- **Applicable to all industries**, and to organisations of any size. It describes *what* must be in place, never *how* to do it.
- Focused on meeting **customer and regulatory requirements**.

WHY IT SAYS WHAT, NOT HOW

The same clause must work for a shipyard and a dental practice. So it requires “documented information to the extent necessary” rather than “a manual of at least 40 pages”. That generality is deliberate — and it is also why the standard reads as abstractly as it does.

THE CONSEQUENCE FOR YOU

A requirement you cannot picture in a real organisation is a requirement you have not understood yet. Throughout the next section, keep asking: *what would I physically see if I walked in?*

What the certificate actually says

The most important misunderstanding in this course

What it certifies

That the organisation has a **quality management system** meeting the requirements of ISO 9001 — processes defined, controlled, measured and improved.

What it does *not* certify

That the **product** is good. ISO 9001 is a system standard, not a product standard. It says nothing about how well the product performs.

THE UNCOMFORTABLE CONSEQUENCE

A company can be fully ISO 9001 certified and make a product you would not buy — provided it makes it *consistently*, to its own declared specification, and improves the process over time. Certification attests to control and repeatability, not excellence.

SO WHY DOES ANYONE WANT IT?

Because large customers and regulated industries require it before they will place an order. It is a licence to bid. Recall the New Approach: the law points at standards, and contracts rely on them.

The structure of ISO 9001

Ten clauses — the first three are preliminaries, the work is in 4 to 10

1 Scope • **2** Normative references • **3** Terms and definitions — these three contain no requirements

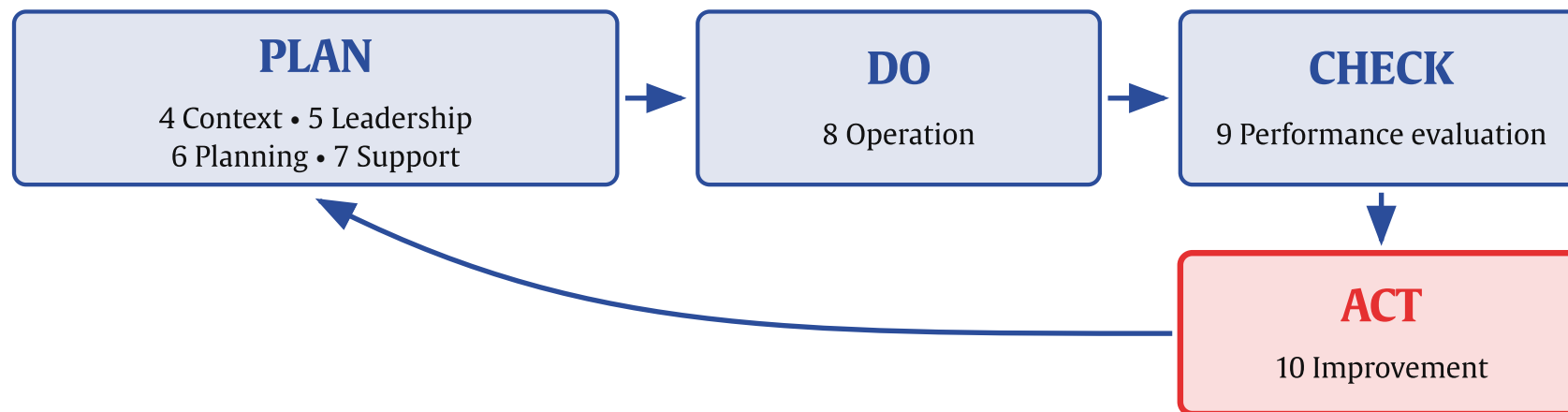
- **4 — Context of the organization** who you are, what you do, and what your interested parties expect
- **5 — Leadership** management's role in driving the system
- **6 — Planning** setting objectives and identifying risks and opportunities
- **7 — Support** the resources you need: people, infrastructure, documented information
- **8 — Operation** actually producing the product or delivering the service
- **9 — Performance evaluation** measuring how well you are doing
- **10 — Improvement** corrective action and continual improvement

This structure is not unique to quality

Clauses 4–10 are the *High Level Structure* shared by all modern ISO management system standards. Learn it once, and you can read any of them.

The structure is PDCA

The only thing you need to memorise



WHY THIS IS WORTH MEMORISING

If you can place a requirement in the right clause, you can usually reconstruct what it says. An auditor finding that objectives were never reviewed is looking at clause 9, not clause 6 — the objective was set, it was the *checking* that failed.

Recap **TAKEAWAYS**

Before we go inside the clauses



- **1** **ISO 9001 says *what*, never *how*** — because the same clause must work for a shipyard and a dental practice.
- **2** **A certificate attests to the system, not the product.** Control and repeatability, not excellence.
- **3** **Ten clauses; 1–3 are preliminaries.** The requirements live in 4 to 10.
- **4** **Clauses 4–10 are PDCA** — plan (4,5,6,7), do (8), check (9), act (10). Memorise this and you can reconstruct the rest.

PART 4

Inside the clauses

*What each one actually asks for, and what you would
see if you walked in*

Clause 4 — Context of the organization

PLAN — new in 2015

- **Understand the context** — the internal and external issues that affect your ability to deliver.
- **Identify interested parties** — who they are and what they require. Customers, yes, but also regulators, owners, employees, neighbours.
- **Determine the scope** — which products, sites and activities the QMS covers. This is what the certificate will say.
- **Define the processes** and how they interact.

SCOPE IS WHERE CERTIFICATES GET QUIETLY NARROW

A company may be certified for “design and manufacture of valves at the Perugia site” — not for its second factory, and not for its service division. A customer who reads only the logo and not the scope line has learned less than they think.

Clause 5 — Leadership

PLAN

- **Leadership and commitment** — top management must take accountability for the effectiveness of the QMS. Not sponsor it. *Be accountable for it.*
- **Quality policy** — a short statement of intent, appropriate to the organisation, communicated and understood.
- **Roles, responsibilities and authorities** — assigned and communicated, so that everyone knows who decides what.

WHAT A BAD QUALITY POLICY LOOKS LIKE

“We are committed to excellence and total customer satisfaction.” True of every organisation on earth, therefore useless. A policy that could be swapped with a competitor’s without anybody noticing is decoration, not direction.

TRY IT

Write a one-sentence quality policy for this course that a competitor could *not* also use.

Clause 6 — Planning

PLAN — where risk-based thinking lives

- **Actions to address risks and opportunities** — identify what could go wrong, and what could go unexpectedly right, and plan for both.
- **Quality objectives** — measurable, monitored, communicated, and updated. With a plan saying what will be done, by whom, by when, and how results will be evaluated.
- **Planning of changes** — changes to the QMS are made deliberately, not accidentally.

“MEASURABLE” IS DOING REAL WORK IN THAT SENTENCE

“Improve customer satisfaction” is not an objective. “Reduce late deliveries from 12% to 5% by 30 June, owned by the logistics manager, reviewed monthly” is. The standard asks for the second, and this is the single most common finding in a real audit.

Clause 7 — Support

PLAN — people, infrastructure, knowledge, documents

- **Resources** — people, infrastructure, and a suitable working environment.
- **Competence** — determine what competence is needed, ensure people have it, and *retain evidence*. Training records exist for this clause.
- **Awareness** — people must know the quality policy, the objectives relevant to them, and what their contribution means.
- **Organizational knowledge** — the knowledge the organisation needs must be maintained, not held in one person's head.
- **Documented information** — created, controlled, versioned, available where needed.

AWARENESS IS NOT A POSTER

An auditor tests this clause by asking an operator, not a manager: *what does the quality policy mean for your job?* If the answer is a blank look, the poster on the wall has not satisfied the requirement.

Clause 7 — Monitoring and measuring resources

Where this course's two halves meet

Where measurement is used to verify conformity, the equipment must be **suitable, maintained**, and — where measurement traceability is required — **calibrated against national or international measurement standards**, with the basis for calibration retained.

Follow the chain upward

Your gauge is calibrated against a reference. That reference is calibrated by an accredited laboratory. That laboratory is accredited by **ACCREDIA**. ACCREDIA is recognised internationally. Only then does your measurement mean the same thing in Perugia and in Hamburg.

WHY AN UNCALIBRATED GAUGE IS WORSE THAN NONE

With no gauge you know you are guessing. With an uncalibrated gauge you have records, confidence, and a signed release certificate — all of them wrong, and all of them documented. The second situation is far more expensive when it unwinds.

This is the bridge

Everything from Lecture 7 onwards — statistics, measurement techniques, uncertainty — exists to satisfy this one sub-clause properly.

Clause 8 — Operation

DO — the longest clause, because it is the actual work

- **Operational planning and control** — plan the processes needed, with criteria and resources.
- **Requirements for products and services** — communicate with customers, determine requirements, and *review them before committing*.
- **Design and development** — inputs, controls, outputs, reviews, verification, validation and change control.
- **Externally provided processes** — control your suppliers and anything you outsource.
- **Production and service provision** — controlled conditions, identification and traceability, preservation, post-delivery activities.
- **Control of nonconforming outputs** — identify them, and stop them being used by accident.

“REVIEW BEFORE COMMITTING” PREVENTS THE CLASSIC DISASTER

A salesperson promises a delivery date and a tolerance the factory cannot meet. The requirement was accepted before anyone competent looked at it. This clause exists because that happens constantly.

Clause 9 — Performance evaluation

CHECK

- **Monitoring, measurement, analysis and evaluation** — decide what to measure, when, and how the results are evaluated. Including **customer satisfaction**.
- **Internal audit** — at planned intervals, the organisation audits *itself* against the standard and against its own arrangements.
- **Management review** — top management reviews the system at planned intervals, considering audit results, customer feedback, objectives, risks and opportunities for improvement.

THE INTERNAL AUDIT IS THE SYSTEM'S IMMUNE RESPONSE

A company that only finds problems when the external auditor arrives has a system that works once a year. The internal audit exists so that non-conformities are found by people who want to fix them, not by someone deciding whether to withdraw your certificate.

Clause 10 — Improvement

ACT

- **Nonconformity and corrective action** — react to it, deal with the consequences, then **evaluate the need to eliminate the cause so it does not recur**.
- **Continual improvement** — improve the suitability, adequacy and effectiveness of the QMS on an ongoing basis.

Correction is not corrective action

Correction fixes the thing: scrap the part, reprint the label, apologise to the customer. **Corrective action** removes the cause so the next one never happens. Almost every organisation does the first and reports it as the second.

A TEST YOU CAN APPLY IN ANY WORKPLACE

Ask: “has this problem happened before?” If yes, then whatever was done last time was a correction, not a corrective action — however it was written up. Recurrence is the proof.

- **1 4 Context** — interested parties and the *scope*, which is what the certificate actually covers.
- **2 5 Leadership** — accountability at the top, and a quality policy a competitor could not reuse.
- **3 6 Planning** — risks and opportunities, plus objectives that are genuinely measurable.
- **4 7 Support** — competence, awareness, documented information, and calibrated measuring equipment traceable to national standards.
- **5 8 Operation** — review requirements before committing; control suppliers; quarantine nonconforming output.
- **6 9 & 10** — internal audit and management review; and corrective action means removing the *cause*, not just the symptom.

What the 2015 revision changed

- **Risk-based thinking** — the headline change. Identify risks and opportunities that could affect your processes, and plan for them.
- **Leadership, not a representative** — earlier versions let a company appoint a “management representative” to own quality. 2015 removed that and put the duty on top management directly.
- **Context of the organization** — a new clause 4. You must identify your interested parties and what they require, rather than assuming.
- **Documented information** — replaced the rigid prescription of manuals and procedures. What matters is that the information exists and is controlled, not that it is bound in a book.

RISK-BASED THINKING REPLACED “PREVENTIVE ACTION”

Before 2015 there was a separate clause requiring preventive action — which in practice meant a form filled in after something had already gone wrong. Making risk part of planning was the attempt to make prevention real rather than ceremonial.

Learn it once, read them all

The High Level Structure across management system standards

ISO 9001

Quality management. Interested party: the customer.

ISO 14001

Environmental management. Interested party: the regulator, the neighbourhood, the planet.

ISO 45001

Occupational health and safety. Interested party: the worker.

All three share clauses 4 to 10, the same wording for context, leadership, planning, support, operation, evaluation and improvement. Only the subject matter changes.

WHY THIS WAS DONE DELIBERATELY

Before the common structure, a company holding three certificates ran three parallel systems with three sets of audits and three manuals. Now one management review, one internal audit programme and one document system can serve all three. The standard was restructured to reduce its own burden.

Benefits — and the honest limits

What certification buys you

- improved quality and operational efficiency
- recognition and trust in global markets, where it is often demanded by large clients or in regulated industries
- risk management — addressing issues before they become compliance problems
- a structure that supports long-term improvement

What it does not buy you

- a good product — the system is certified, not the output
- immunity from the bureaucratic degeneration we saw in TQC
- engagement, if the organisation treats the audit as the goal

The test of a real system

Ask what happens the week *after* the auditor leaves. If the behaviour changes, the company has a certificate. If it does not, the company has a quality system.

Put it to work

Apply the clauses to something you know

TAKE THIS UNIVERSITY AS THE ORGANISATION

1. **Clause 4** — who are the interested parties, and what does each require? Where do those requirements conflict?
2. **Clause 6** — write one genuinely measurable quality objective for a degree programme.
3. **Clause 7** — what is the equivalent of “calibrated measuring equipment” when the output is a graduate?
4. **Clause 10** — name a recurring problem here. Was the response a correction or a corrective action?

QUESTION 3 IS THE ONE WORTH ARGUING ABOUT

If the exam is the measuring instrument, what would it mean for it to be calibrated — and traceable to what reference? Every accreditation and quality-assurance system in higher education is an attempt to answer exactly this.

Where we got to **TAKEAWAYS**

- **1** **ISO 9000** is vocabulary and principles; **ISO 9001** is requirements. Only 9001 is certifiable.
- **2** **Seven principles**, all of which you had already met under other names from Deming, Juran and Ishikawa.
- **3** **Clauses 4–10 are PDCA** — and the same structure carries ISO 14001 and ISO 45001.

Where we got to **TAKEAWAYS**

- **1 A certificate attests to a system, not a product** — and only within the scope printed on it.
- **2 Clause 7 is the bridge to the second half of this course:** measurement traceable to national standards, which is why metrology follows.
- **3 Corrective action removes the cause.** If the problem recurs, what you did was a correction.

Next time

Implementing it: gap analysis, process mapping, the certification audit itself — and what happens when it goes wrong.

Questions?

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